

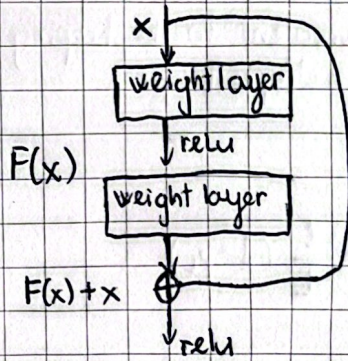
Section 16: CNN Architectures

1 CNN architectures:

- + Alexnet
- + VGGnet
- + Inception Net (Google net)

+ Network-in-network

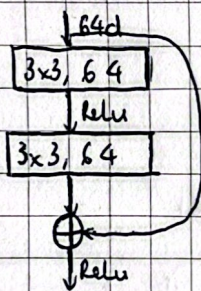
2 Residual Networks



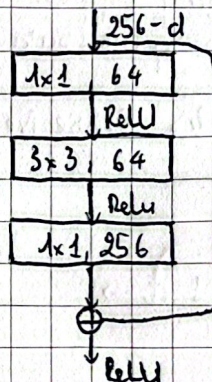
We are learning what's left of (residual) without learning a new represent. If the block doesn't learn't any thing, then your $F(x)$ would be 0, and that is what makes the training go much faster

There are two types

Basic



Bottle neck



• Is it better to put batch normalization after the addition or before the addition at the end of the residual block?

→ before the addition

• Residual Networks don't use Max pooling to down-sample the size of feature maps. Instead, they use a stride of 2.

3. Stochastic Depth

Are deeper layers get better accuracy? NO

Deeper layers causes over fitting and gradient vanishing

Stochastic Depth Regularization aims to shrink the depth of a network during training, while keeping it unchanged during testing. Train short networks and use deep networks at test time.

For each mini-batch we randomly select sets of blocks and remove their corresponding transformation functions (block), only keeping the identity (or skip connection).



The survival probability p_i for a block i , equivalent to the keeping probability in dropout.

For the last block L , $p_L = 0.5$.

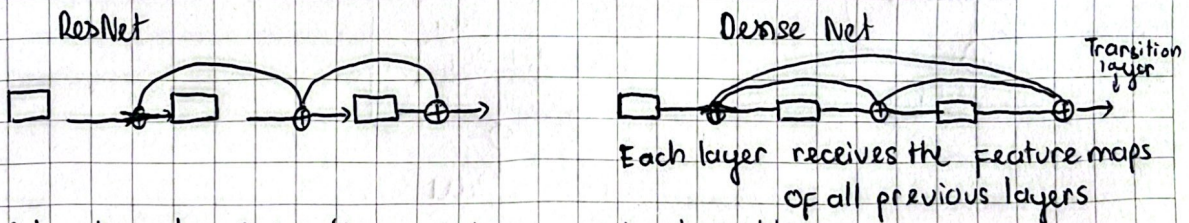
p_i is decayed linearly from $p_0 = 1$, $p_L = 1 - \frac{L}{L} (1 - p_L)$

During testing, multiply the block by p_i .

4. More CNN architectures

- + ResNeXt: Combined InceptionNet with ResNet
- + Efficient Net: Scaling different dimensions (depth, width and resolution) together (balancing)

5. Densely Connected Networks (Dense Net)



Advantage 1: Solve the vanishing gradient problem

Advantage 2: Feature preserving and avoiding redundancy

At the end of each dense block, the number of feature maps get larger.

Transition layer is added, which consist of:

- 1x1 conv to half.
- 2x2 avg pool.

6. Squeeze-Excite Networks (SENet)

7. Seperable Convolutions

8. Transfer learning

Three ways of transfer learning.

1) Fine tune the whole network.

2) Freeze first-part of the network and fine-tuning the last layer of the network.

3) Freeze first-part of the network and train the last layer from scratch.

9. Is a 1×1 convolutional filter equivalent to a FC layer?